

1. Evaluate the following limits. Simplify your answers where possible.

a. $\lim_{x \rightarrow 27} \frac{x+27}{\sqrt[3]{x}+3}$

let $t = \sqrt[3]{x}$ $t \rightarrow 3$
 $t^3 = x$

$$\lim_{t \rightarrow 3} \frac{t^3 + 27}{t + 3}$$

$$= \lim_{t \rightarrow 3} \frac{(t+3)(t^2 - 3t + 9)}{t+3}$$

$$= 9$$

b. $\lim_{x \rightarrow 2} \frac{\frac{1}{x} - \frac{1}{2}}{x-2}$

$$= \lim_{x \rightarrow 2} \frac{\frac{2-x}{2x}}{x-2}$$

$$= \lim_{x \rightarrow 2} \frac{-1}{2x}$$

$$= -\frac{1}{4}$$

c. $\lim_{x \rightarrow 0} \frac{(2+x)^3 - 8}{x}$

$$= \lim_{x \rightarrow 0} \frac{(2+x-2)((2+x)^2 + 2(2+x) + 4)}{x}$$

$$= \lim_{x \rightarrow 0} (2+x)^2 + 2(2+x) + 4$$

$$= 12$$

d. $\lim_{t \rightarrow 0} \frac{t}{\sqrt{1+3t} - 1}$

$$= \lim_{t \rightarrow 0} \frac{t(\sqrt{1+3t} + 1)}{(\sqrt{1+3t} - 1)(\sqrt{1+3t} + 1)}$$

$$= \lim_{t \rightarrow 0} \frac{t(\sqrt{1+3t} + 1)}{t + 3t}$$

$$= \lim_{t \rightarrow 0} \frac{\sqrt{1+3t} + 1}{3}$$

$$= \frac{2}{3}$$

e. $\lim_{t \rightarrow 0} \left(\frac{1}{t\sqrt{t+1}} - \frac{1}{t} \right)$

$$= \lim_{t \rightarrow 0} \frac{1}{t} \left[\frac{1}{\sqrt{t+1}} - 1 \right]$$

$$= \lim_{t \rightarrow 0} \frac{1}{t} \left[\frac{1 - \sqrt{t+1}}{\sqrt{t+1}} \right]$$

$$= \lim_{t \rightarrow 0} \frac{1}{t} \left[\frac{1 - \sqrt{t+1}}{\sqrt{t+1}} \right] \left[\frac{1 + \sqrt{t+1}}{1 + \sqrt{t+1}} \right]$$

$$= \lim_{t \rightarrow 0} \frac{-t}{(t\sqrt{t+1})(1 + \sqrt{t+1})} = -\frac{1}{2}$$

f. $\lim_{x \rightarrow 1} \frac{\sqrt[9]{x} - 1}{\sqrt[3]{x} - 1}$

let $t = \sqrt[9]{x}$
 $t^3 = \sqrt[3]{x}$
 $t \rightarrow 1$

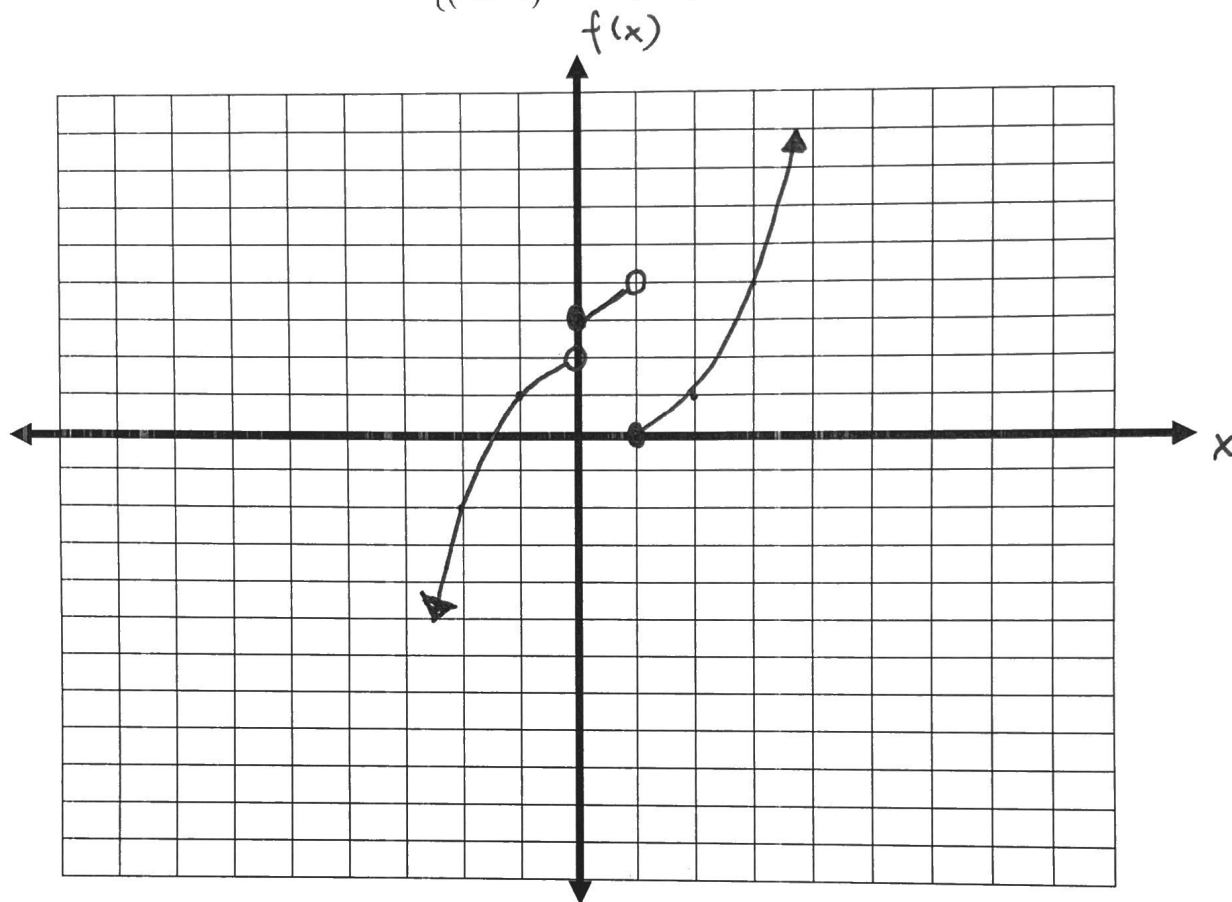
$$\lim_{t \rightarrow 1} \frac{t-1}{t^3-1}$$

$$= \lim_{t \rightarrow 1} \frac{t-1}{(t-1)(t^2+t+1)}$$

$$= \frac{1}{3}$$

2. Sketch a properly labelled graph of the following piecewise function.

$$f(x) = \begin{cases} 2-x^2 & , x \in (-\infty, 0) \\ x+3 & , x \in [0, 1) \\ (-x+1)^2 & , x \in [1, \infty) \end{cases}$$



3. Determine whether or not the following function is continuous on the interval $(-\infty, \infty)$.
If the function is not continuous, state all value(s) of x where the function is discontinuous.

$$f(x) = \begin{cases} x^3 - 1 & , x \in (-\infty, 0) \\ -(4x+1) & , x \in [0, 1) \\ 2\sqrt{x} - 7 & , x \in [1, \infty) \end{cases}$$

@ x = 0

① $f(0) = -1$

② $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x^3 - 1) = -1$
 $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} -(4x+1) = -1$

$\lim_{x \rightarrow 0} f(x) = -1$

③ $f(0) = \lim_{x \rightarrow 0} f(x) = -1$

∴ cont. @ $x = 0$

@ x = 1

① $f(1) = -5$

② $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} -(4x+1) = -5$
 $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (2\sqrt{x} - 7) = -5$

$\lim_{x \rightarrow 1} f(x) = -5$

③ $f(1) = \lim_{x \rightarrow 1} f(x) = -5$ $x = 1$

∴ cont. at $x = 1$

4. Determine $\lim_{x \rightarrow 3} \frac{\sqrt{7-x}-2}{1-\sqrt{4-x}}$.

$$= \lim_{x \rightarrow 3} \frac{(\sqrt{7-x}-2)(\sqrt{7-x}+2)(1+\sqrt{4-x})}{(1-\sqrt{4-x})(\sqrt{7-x}+2)(1+\sqrt{4-x})}$$

$$= \lim_{x \rightarrow 3} \frac{(7-x-4)(1+\sqrt{4-x})}{(1-4+x)(\sqrt{7-x}+2)}$$

$$= \lim_{x \rightarrow 3} \frac{(\cancel{3-x})(1+\sqrt{4-x})}{(\cancel{x-3})(\sqrt{7-x}+2)}$$

$$= \lim_{x \rightarrow 3} \frac{-(1+\sqrt{4-x})}{\sqrt{7-x}+2}$$

$$= -\frac{1}{2}$$

5. Determine all values of b and c in the function below, where $b, c \in \mathbb{R}$ such that $f(x)$ is continuous on the interval $(-\infty, \infty)$. Show all steps.

$$f(x) = \begin{cases} cx-2 & , x \in (-\infty, -1) \\ -bx^2+2c & , x \in [-1, 1) \\ c\sqrt{x}+b+1 & , x \in [1, \infty) \end{cases}$$

@ $x = -1$

① $f(-1) = -b+2c$

② $\lim_{x \rightarrow -1^-} (cx-2) \begin{cases} \lim_{x \rightarrow -1^+} -bx^2+2c \\ = -b+2c \end{cases}$
 $= -c-2$

$-c-2 = -b+2c$

$\boxed{b-3c=2} \quad (1)$

② - 2 × ①

$2b-c = -1$
 $-2(b-3c=2)$

$5c = -5 \Rightarrow \boxed{c = -1}$

@ $x = 1$

① $f(1) = c+b+1$

② $\lim_{x \rightarrow 1^-} -bx^2+2c \begin{cases} \lim_{x \rightarrow 1^+} c\sqrt{x}+b+1 \\ = c+b+1 \end{cases}$
 $= -b+2c$

$-b+2c = c+b+1$

$\boxed{2b-c=-1} \quad (2)$

Sub. $c = -1$ into ①

$b+3=2$

$\boxed{b = -1}$

4. Determine $\lim_{x \rightarrow 3} \frac{\sqrt{7-x}-2}{1-\sqrt{4-x}}$.

$$= \lim_{x \rightarrow 3} \frac{(\sqrt{7-x}-2)(\sqrt{7-x}+2)(1+\sqrt{4-x})}{(1-\sqrt{4-x})(\sqrt{7-x}+2)(1+\sqrt{4-x})}$$

$$= \lim_{x \rightarrow 3} \frac{(7-x-4)(1+\sqrt{4-x})}{(1-4+x)(\sqrt{7-x}+2)}$$

$$= \lim_{x \rightarrow 3} \frac{(\cancel{3-x})(1+\sqrt{4-x})}{(\cancel{x-3})(\sqrt{7-x}+2)}$$

$$= \lim_{x \rightarrow 3} \frac{-(1+\sqrt{4-x})}{\sqrt{7-x}+2}$$

$$= -\frac{1}{2}$$

5. Determine all values of b and c in the function below, where $b, c \in \mathbb{R}$ such that $f(x)$ is continuous on the interval $(-\infty, \infty)$. Show all steps.

$$f(x) = \begin{cases} cx-2 & , x \in (-\infty, -1) \\ -bx^2+2c & , x \in [-1, 1) \\ c\sqrt{x}+b+1 & , x \in [1, \infty) \end{cases}$$

@ $x = -1$

① $f(-1) = -b+2c$

② $\lim_{x \rightarrow -1^-} (cx-2) \begin{cases} \lim_{x \rightarrow -1^+} -bx^2+2c \\ = -b+2c \end{cases}$
 $= -c-2$

$-c-2 = -b+2c$

$\boxed{b-3c=2} \quad (1)$

② $- 2 \times (1)$

$2b-c=-1$
 $-2(b-3c=2)$

$5c = -5 \Rightarrow \boxed{c = -1}$

@ $x = 1$

① $f(1) = c+b+1$

② $\lim_{x \rightarrow 1^-} -bx^2+2c \begin{cases} \lim_{x \rightarrow 1^+} c\sqrt{x}+b+1 \\ = c+b+1 \end{cases}$
 $= -b+2c$

$-b+2c = c+b+1$

$\boxed{2b-c=-1} \quad (2)$

Sub. $c = -1$ into ①

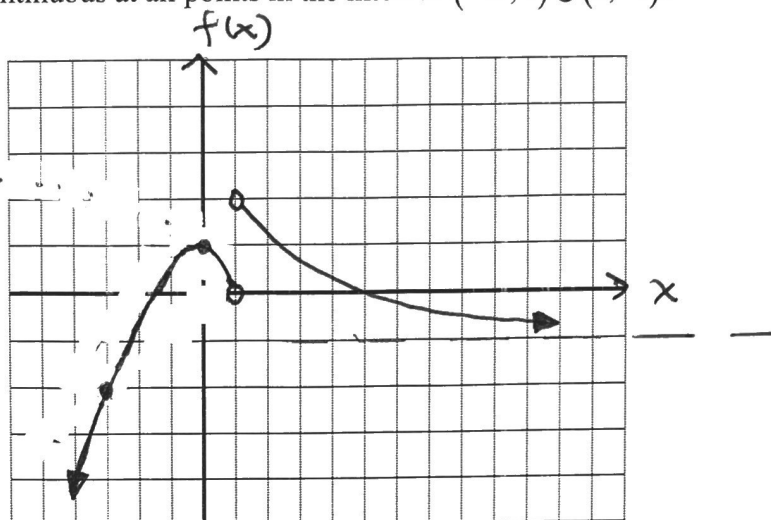
$b+3=2$

$\boxed{b = -1}$

6. Sketch the graph of **any** function that satisfies the following conditions:

$$f(0)=1, \quad \lim_{x \rightarrow -3} f(x) = -2, \quad \lim_{x \rightarrow 1^-} f(x) = 0, \quad \lim_{x \rightarrow 1^+} f(x) = 2, \quad \lim_{x \rightarrow \infty} f(x) = -1$$

The function is continuous at all points in the interval $(-\infty, 1) \cup (1, \infty)$.



7. If $\lim_{x \rightarrow -1} \frac{x^3 - ax^2 - x + 4}{x+1}$ exists, determine the value of a .

$$f(-1) = (-1)^3 - a(-1)^2 - (-1) + 4 = 0$$

$$-1 - a + 1 + 4 = 0$$

$$-a + 4 = 0$$

$$a = 4$$

8. Determine **all** values of p and q such that $\lim_{x \rightarrow 0} \frac{\sqrt{px+q}-3}{x} = 1$, where $p, q \in \mathbb{R}$.

$$f(x) = \sqrt{px+q} - 3 \quad \lim_{x \rightarrow 0} \frac{(\sqrt{px+q}-3)(\sqrt{px+q}+3)}{x(\sqrt{px+q}+3)}$$

$$f(0) = 0$$

$$\sqrt{q} - 3 = 0$$

$$\boxed{q = 9}$$

$$= \lim_{x \rightarrow 0} \frac{\cancel{px+q}-9}{x(\sqrt{px+q}+3)}$$

$$= \lim_{x \rightarrow 0} \frac{p}{\sqrt{px+q}+3}$$

$$= \frac{p}{6} = 1 \Rightarrow \boxed{p = 6}$$